

RESEARCH PAPER

Impact of value co-creation in the artificial intelligence innovation ecosystem on competitive advantage and innovation intelligibility

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Abstract

Artificial intelligence has been widely adopted for digital transformation and value creation for enterprises. However, studies about how participants can co-create value in the AI innovation ecosystem are needed. Thus, this research proposed a comprehensive research model to examine the impact of value co-creation in the AI innovation ecosystem for competitive advantage and innovation intelligibility with the mediating effect of dynamic capabilities and innovation capabilities. We collected data from 234 AI-related manufacturers for the structural equation modelling analysis. Results show that (1) the AI innovation ecosystem can bring economic value and relationship value to enterprises; (2) impacts of economic value and relationship value on competitive advantage and innovation intelligibility are not significant; and (3) value co-creation does not have a significant impact on competitive advantage and innovation intelligibility, and dynamic capabilities and innovation capabilities play the mediating role. Theoretical contribution, practical implication, limitation, and future research are presented.

KEYWORDS

AI innovation ecosystem, competitive advantage, innovation intelligibility, value co-creation

1 | INTRODUCTION

Artificial intelligence (AI) refers to intelligence demonstrated by machines (Broo et al., 2022; Javaid et al., 2022; Lu, 2019; Xu, 2021; Zhang & Lu, 2021). AI has been widely adopted in different industries (Chen et al., 2021, 2022; Lu & Ning, 2020; Norinder & Norinder, 2022). The open innovation of the R&D process has been shifted to the value co-creation of the innovation ecosystem to improve innovation intelligibility. Manufacturing enterprises in China face internal and external challenges such as rising labour costs,

strict environmental regulations, and COVID-19. Thus, it has been proposed to utilize such technologies as the Internet, big data, and artificial intelligence to support the optimization, upgrading, and digitization of traditional manufacturing industries. Although multinational companies have achieved a certain degree of digitization, small and medium-sized enterprises (SME) lack sufficient resources and capabilities for their digital transformation. In this context, proposing the value co-creation of the AI innovation ecosystem provides new ideas and new methods to achieve artificial intelligibility.

Value co-creation has recently attracted research attention (Bai, 2018; Furtado et al., 2017; Rasool et al., 2017). However, most scholars studied the value co-creation behaviour from different perspectives, including digital interactive platforms such as social media, virtual communities, platform-based e-commerce, and shared economy platforms (Bu et al., 2016; Chen et al., 2018; Fan et al., 2020; Li, 2019; Ramaswamy & Ozcan, 2018, 2019; Shaheer & Li, 2020; Zhou et al., 2018; Zhu et al., 2017), the innovation ecosystem (Dai et al., 2018; Ketonen-Oksi & Valkokari, 2019; Zhou et al., 2019), the business ecosystem (Hu & Lu, 2018; Wang & Li, 2018), the service ecosystem (Jian et al., 2016; Linghu et al., 2019; Meynhardt et al., 2016), and other factors and mechanisms of value co-creation behaviour (Chi et al., 2020; Jiang & Li, 2016; Meng, 2019; Sun et al., 2019; Wang & Zhu, 2018; Wang & Zhu, 2019; Xiao & Ma, 2018; Zhang et al., 2020). In addition, researchers have discussed the impact of value co-creation on the competitive advantage or enterprises' artificial intelligibility (e.g. Wang & Luan, 2019; Zhou, 2015; Zhu et al., 2019). However, most studies investigated the intermediary effect of value co-creation. Therefore, the systematic and empirical analysis of the spillover effect of value co-creation is missing.

In this paper, based on previous studies, a comprehensive analysis of the value of the associated ecosystem to create value innovation is conducted, exploring create value, the correlation value, and competitive advantage between the companies. This paper selects a certain number of small and medium-sized manufacturing companies to conduct empirical research, hoping that the research results can provide reference and guidance for small and medium-sized manufacturing companies to participate in the value co-creation of the innovation ecosystem and solve the resource constraint dilemma in the process of innovation and transformation.

2 | LITERATURE REVIEW

2.1 | Value co-creation of the AI innovation ecosystem

The AI innovation ecosystem can be considered as a network of interdependent participants for AI development. The network participants combine proprietary and complementary resources and capabilities to create a primary AI value proposition and pass it on to end-users (Kühl et al., 2020; Walrave et al., 2018). Value co-creation has been extensively studied after a shift from sales to service dominance was proposed by Vargo and Lusch (2004). The service-dominant logic states that value is usually

co-created among multiple participants, including beneficiaries (Sinkovics et al., 2018; Vargo & Lusch, 2016). Participants can create value in an interactive environment (Ramaswamy & Ozcan, 2018). Furthermore, participants in the AI innovation ecosystem can integrate complementary resources through open collaboration, improve the AI innovation efficiency and effectiveness of the AI innovation ecosystem, and ultimately realize value co-creation (Ceccagnoli et al., 2012; Köhl et al., 2020; Zhang, Jiang, et al., 2015a). The value co-creation of the AI innovation ecosystem refers to the open collaboration and resource sharing between entities in AI innovation activities (Iansiti & Levien, 2004; Sarker et al., 2012; Vargo & Lusch, 2008, 2016). The AI innovation ecosystem can help SMEs solve the bottleneck of insufficient resources and promote their AI innovation and development (Ercan & Samet, 2020; Han et al., 2012).

Large enterprises have an important position in the AI innovation ecosystem in the digital era due to their resource advantages (Tilson et al., 2010). The vital resources of large enterprises can provide fertile soil to SMEs' AI innovation and development because SMEs lack resources (Ceccagnoli et al., 2012). Large enterprises and SMEs will generate the two-way flow of knowledge in open collaboration. Additionally, other entities such as investment institutions, scientific research institutions, service institutions, government agencies, and so on, in the AI innovation ecosystem will also provide resources and support (Adner & Kapoor, 2010; Guo et al., 2020). All participants achieve their own goals by supporting the AI innovation and development of innovative entities. Consequently, it will bring value to all participants involved.

The value co-creation model of the AI innovation ecosystem covers many different aspects, including new products or services design and development (Lemke et al., 2011; Mahr et al., 2014), user participation in innovation (Bogers et al., 2010; Hu & Han, 2018), the role of users in customized production (Syam & Pazgal, 2013), co-production (Ranjan & Read, 2016; Storbacka et al., 2016), value co-creation between enterprises and customers (Bettencourt et al., 2014; Cossío-Silva et al., 2016; Xie et al., 2016), value co-creation between suppliers (Sjödin, 2019; Zhang, Jiang, et al., 2015a), value co-creation between employees and suppliers (Ramaswamy & Ozcan, 2018), value co-creation with third party logistics, inter-organizational partnership (Grover & Kohli, 2012), and so on.

2.2 | Spillover effect of value co-creation

In the process of value co-creation, all stakeholders (e.g. suppliers and users) will form a new combination

of resources and capabilities called associated value for all parties (Zhang, Jiang, et al., 2015a). On the one hand, researchers have pointed out that the associated value may be the direct value created for the enterprise or the customer (Ramaswamy & Ozcan, 2018; Xie et al., 2016). Therefore, the associated value is considered as the explicit correlation value in this paper, including economic value and relationship value. Economic value refers to the benefits and costs derived from better and customized products and services (Yim et al., 2012). Relationship value refers to the emotional or relationship bond between the enterprise and the customer (Yim et al., 2012). On the other hand, value co-creation can also bring indirect value to enterprises or customers, such as capabilities improvement (Sjödin, 2019). The indirect value is called implicitly associated value which includes dynamic ability and innovation capabilities.

Value creation and value acquisition are entities' important activities. Value creation is the process of interaction and collaboration between entities at the ecosystem level to create value. Value acquisition is the process of obtaining benefits and building competitive advantages at the enterprise level (Ritala et al., 2013). The research focuses on the AI innovation ecosystem's value creation and acquisition. Specifically, all participants in the ecosystem obtain value through value co-creation, resulting in competitive advantage and innovation intelligibility.

3 | THEORETICAL FRAMEWORKS

3.1 | Value co-creation and associated value

Value co-creation can bring value to all parties involved in the AI innovation ecosystem (Zhang, Jiang, et al., 2015a). Recent studies have confirmed that both economic value and relationship value of co-creation can be generated through customer participation (Chan et al., 2010; Chen & Wang, 2016; Yim et al., 2012). Customer participation can bring economic value to customers by producing high-quality and customized products/services (Yim et al., 2012). Meanwhile, customers' participation in the value co-creation will have a strong collaboration, resulting in a closer and more beneficial relationship between the company and the customer (Payne et al., 2008; Xie et al., 2016; Yim et al., 2012).

The sharing of internal and external knowledge through formal or informal mechanisms in the AI innovation ecosystem helps enterprises develop new ideas

and capabilities, strengthening the innovation process (Galati & Bigliardi, 2019; Ritala et al., 2018). In a similar vein, enterprises will better utilize internal and external resources to reduce risks and shorten product development cycles for flexibility, market adaptability, and profit channels (Chung & Kim, 2003). Therefore, such opportunities and capabilities have significantly increased the benefits of innovation (Chesbrough, 2007). Enterprises supporting innovation obtain economic returns by participating in AI innovation activities (Oh et al., 2016). Accordingly, this research proposes the following hypotheses:

Hypothesis 1. Value co-creation positively affects economic value.

Hypothesis 2. Value co-creation positively affects relationship value.

Teece et al. (1997) define dynamic capabilities as an enterprise's integration (Gorkhali & Xu, 2016, 2019; Uysal & Mergen, 2021; Xu et al., 2008), establishment, and reconstruction of resources and capabilities to adapt to the ever-changing environment. Ferreira et al. (2020) believe that dynamic capabilities can perceive opportunities and threats, implement strategic decisions and changes, and solve problems. The underlying processes of dynamic capabilities include perception, coordination, learning, integration, and reconstruction (Mikalef & Pateli, 2017). Manufacturers and their logistics companies will collaborate to solve lean and on-time delivery issues. The third-party logistics companies provide more professional technology and equipment, innovate in distribution and logistics services, and assist manufacturers in optimizing their order management process. Thus, the enterprises' dynamic ability will be improved to adapt to the external environment and solve problems.

The value co-creation of the AI innovation ecosystem includes resource sharing and open collaboration. Studies have shown that resource sharing between companies can improve innovation capabilities (Moore, 1993). The cooperation between companies, suppliers, and other partners can promote innovation capabilities (Mina et al., 2014; Zhang et al., 2014). The integration of an enterprise's internal and external knowledge includes the acquisition of external knowledge and transfers internal knowledge to the external environment (Papa et al., 2020; Xu, 2011, 2013, 2014). This mechanism mingles the enterprise with its environment to promote the enterprise's innovation capabilities (Bican et al., 2017). Therefore, this research proposes the following hypotheses:

Hypothesis 3. Value co-creation positively affects dynamic capabilities.

Hypothesis 4. Value co-creation positively affects innovation capabilities.

3.2 | Associated value and competitive advantage

An enterprise has a competitive advantage in creating more value and achieving higher performance than competitors by providing differentiated products/services through cost leadership (Barney, 1991; Lazzarini, 2015; Porter, 1985). Mahdi et al. (2019) believe that maintaining competitive advantage is a dynamic process due to existing and future competitions. Sustained competitive advantage derives from those scarce, valuable, systematic, and un-imitated resources (Barney & Clark, 2007). Obtaining competitive advantages is vital to enterprises during the economic slowdown and intense market competition.

Empirical studies have indicated that customer perceived value (including economic value and relationship value) positively impacts customer satisfaction (Chen, 2008; Chen & Chen, 2010). In the AI innovation ecosystem, the economic value mainly refers to the customized products or services brought by customer participation and the cost reduction brought by the shortened product development cycle. Customization is conducive to improving customer satisfaction and customer loyalty. Low cost improves competitive advantage. Friendly relations can bring added value to customers, and therefore improve customer satisfaction. Different cooperation networks with enterprises, intermediaries, and research institutions positively impact SMEs' competitive advantage because they will obtain the needed resources (Zeng et al., 2010). The relationship value brings closer network relationships. It helps companies make better use of external resources to overcome internal lack of resources and capacity (Nieto & Santamaría, 2007; Sun & Zhao, 2017). Therefore, this research proposes the following hypotheses:

Hypothesis 5a. Economic value positively affects competitive advantage.

Hypothesis 5b. Economic value positively affects innovation intelligibility.

Hypothesis 6a. Relationship value positively affects competitive advantage.

Hypothesis 6b. Relationship value positively affects innovation intelligibility.

Dynamic capabilities are considered as a mechanism to produce competitive advantage and affect innovation intelligibility (Bitencourt et al., 2020). With the rapid development of digitalization, dynamic capabilities based on AI technologies can bring competitive advantage, including the development of new digital capabilities the implementation of organizational changes (Karimi & Walter, 2015; Makkonen et al., 2014; Yang et al., 2019). In the context of supply chain management, Van Poucke et al. (2014) find that dynamic capabilities can affect performance by improving process flexibility and reducing costs. Dynamic capabilities can lead to innovation intelligibility.

Innovation capabilities can create and maintain competitive advantage and innovation intelligibility (Ferreira et al., 2020; Kasa, 2015). In a global market, innovation capabilities are especially critical for companies to gain a competitive advantage (Mazzucchelli et al., 2019). Innovation capabilities mainly include exploration and exploitation that play different roles in innovation. Exploration can usually bring differentiated and innovative output. Exploitation can reduce costs, increase revenue, and improve production efficiency and product quality (O'Cass et al., 2014). In an entrepreneurial-oriented environment, innovation capabilities have a higher value for SMEs that lack resources. SMEs need to constantly seek innovation to maintain their competitive advantage (Ferreira et al., 2020). Innovation capabilities can lead to innovation intelligibility (Brock, 2018). Therefore, this research proposes the following hypotheses:

Hypothesis 7a. Dynamic capabilities positively affect competitive advantage.

Hypothesis 7b. Dynamic capabilities positively affect innovation intelligibility.

Hypothesis 8a. Innovation capabilities positively affect competitive advantage.

Hypothesis 8b. Innovation capabilities positively affect innovation intelligibility.

3.3 | Value co-creation and competitive advantage

Studies have examined customer participation as a competitive advantage (Bendapudi & Leone, 2003; Vargo & Lusch, 2004). By participating in the products or service's

co-production, customers can perceive more value and thereby increase satisfaction (Yim et al., 2012). The co-creation value between enterprises and customers has a significant positive impact on customer loyalty (Cossio-Silva et al., 2016), which in turn brings a competitive advantage.

Entities in the AI innovation ecosystem need to collaborate with other organizations to support innovative activities to improve innovation intelligibility. Such collaborations refer to cooperation with external entities, including customers, suppliers, competitors, universities or other institutions to establish formal or informal connections to improve performance (Ferrerás-Méndez et al., 2015; Jordão & Novas, 2017; Kalyani & Ayachit, 2019; Scuotto et al., 2017; Spigel, 2015). In particular, cross-border AI innovation cooperation has become a strategic source of innovation and competitive advantage. Cooperative companies can develop advanced AI technologies (Mazzucchelli et al., 2019). By acquiring new strategic innovation capabilities and combining internal and external knowledge, cross-border AI innovation cooperation enables companies to improve the quality and efficiency of AI technologies jointly, streamline the time to market for innovative products, and enter into new international markets (Brattström, 2018; Martínez-Noya & Narula, 2018). Customer participation in value co-creation has a significant positive effect on traditional manufacturing product development (Hu & Han, 2018; Kasie et al., 2017). SMEs can create a mechanism to utilize external knowledge for innovation intelligibility (Zhou, 2015). In addition, the value co-creation of manufacturers and third-party logistics can improve customer satisfaction and affect the manufacturers' competitive advantage. Therefore, this research proposes the following hypotheses:

Hypothesis 9. Value co-creation positively affects competitive advantage.

Hypothesis 10. Value co-creation positively affects innovation intelligibility.

Based on the above discussion, the research model with proposed hypotheses is presented in the below Figure 1.

4 | RESEARCH METHODOLOGY

4.1 | Sample selection and data sources

This study collected data from the general manager/CEO of the AI-related manufacturing companies in the Yangtze River Delta region. We sent questionnaires to 400 companies in 2019 and received 234 valid responses, resulting in a 58% response rate. The sample data are summarized in Table 1.

4.2 | Measurement items

The measurement items used in this study are either adopted or adapted from previous studies. We conducted a pilot study using senior executives from well-known enterprises in the Zhejiang Province. The measurement items were revised based on the feedback for content validity. In this study, a 5-point Likert scale was used to measure the variables in the model. The value ranges from 1 (*very inconsistent*) to 5 (*very consistent*).

This study mainly explores the value of co-creation behaviour among the participants in the AI innovation ecosystem. Therefore, this study adopted Ranjan and Read (2016) co-productions 12 items to measure value co-creation. Three items were adopted from Yim et al. (2012) to measure economic value and relationship value, respectively. We adopted five items from Mikalef and Pateli (2017) to measure the dynamic capabilities. As to innovation capabilities, five items from Mazzucchelli

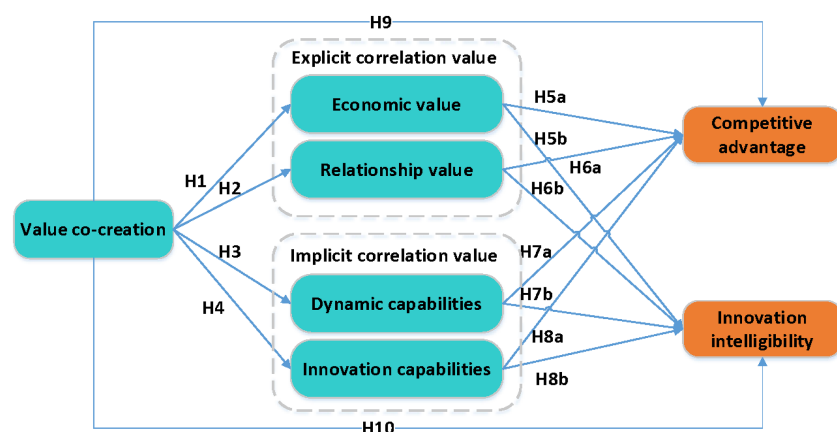


FIGURE 1 Research model with proposed hypotheses [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/tes.2860)]

TABLE 1 Summary of sample data

Classification standard		Sample size	Percentage (%)	Classification standard		Sample size	Percentage (%)
Enterprise nature	State-owned enterprise	89	38.0	Established	<3 years	22	9.4
	Private enterprise	113	48.3		3–5 years	21	9.0
	Joint venture	11	4.7		5–10 years	27	11.5
	Foreign companies	12	5.1		>10 years	164	70.1
	Others	9	3.9	Number of employees	≤20 people	9	3.8
Area	Zhejiang Province	95	40.6		20–300 people	74	31.6
	Jiangsu province	72	30.8		300–1000 people	135	57.7
	Shanghai	56	23.9	Operating income	>1000 people	16	6.9
	Others	11	4.7		<3 million	9	3.8
Industry	Textile and apparel	43	18.4		3–20 million	72	30.8
	Car parts	65	27.8		20–400 million	138	59.0
	Chemical manufacturing	21	10.3		>400 million	15	6.4
	Electrical machinery	28	12.0				
	Electronic equipment	49	20.9				
	Others	28	10.6				

TABLE 2 Variable mean, standard deviation, and correlation coefficient

Variable	Average	Standard deviation	1	2	3	4	5	6
Value co-creation	4.231	0.721	1					
Economic value	4.350	0.739	0.696**	1				
Relationship value	4.062	0.808	0.688**	0.662**	1			
Dynamic capabilities	4.195	0.769	0.590**	0.573**	0.621**	1		
Innovation capability	3.995	0.795	0.585**	0.569**	0.613**	0.695**	1	
Competitive advantage	4.040	0.765	0.675**	0.565**	0.626**	0.693**	0.685**	1
Innovation intelligibility	3.909	0.859	0.661**	0.548**	0.635**	0.697**	0.689**	0.655**

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

et al. (2019) were used in this study. We adapted six items to measure competitive advantage from several studies (Cao et al., 2019; Ferreira et al., 2020; Papadas et al., 2019). We adapted six items from previous studies (Qian et al., 2010) to measure innovation intelligibility.

5 | DATA ANALYSIS AND RESULTS

5.1 | Correlation analysis

The descriptive statistics and correlation analysis results of each variable are shown in Table 2. The correlation

analysis results show that the variables are significantly correlated. Additionally, the correlation coefficients between all variables are less than 0.7, indicating that no serious multicollinearity exists between the variables.

5.2 | Reliability and validity test

Cronbach α coefficient can be used to check reliability that refers to the degree of consistency for all items when measuring the same variable. As shown in Table 3, all variables' coefficients are greater than 0.9. Therefore, the measurement items used in this study have high reliability for subsequent empirical analysis.

Validity refers to the degree to which the scale can accurately measure the variables, including content validity, convergent validity, and discriminant validity. The measurement items were mainly adopted or adapted from previous studies. Therefore, the content validity of the measurement items is relatively high.

As shown in Table 4, the standardized factor loads of all variables are greater than 0.7. The combined

reliability values are greater than 0.9. The average refined variance values are greater than 0.6. Therefore, we observe the measurement items' convergent validity. In addition, the discriminant validity is demonstrated for measurement items because the correlation coefficients between each variable and other variables are less than the square root of the average refined variance of the variable in Table 4.

Variable	Number of items	Item	α coefficient
Value co-creation (VC)	12	VC1-VC12	0.958
Economic value (EV)	3	EV1-EV3	0.920
Relationship value (RV)	3	RV1-RV3	0.916
Dynamic capabilities (DC)	5	DC1-DC5	0.939
Innovation capabilities (IC)	5	IC1-IC5	0.906
Competitive advantage (CA)	6	CA1-CA6	0.927
Innovation intelligibility (II)	6	II1-II6	0.959

TABLE 3 Result of reliability analysis

TABLE 4 Result of convergent and discriminant validity analysis

Variable	Item	Normalized factor loading	Combination reliability	Average refined variance	Square root of average refined variance
Value co-creation (VC)	VC1	0.792	0.958	0.654	0.809
	VC2	0.810			
	VC3	0.832			
	VC4	0.850			
	VC5	0.761			
	VC6	0.819			
	VC7	0.811			
	VC8	0.805			
	VC9	0.820			
	VC10	0.806			
	VC11	0.796			
	VC12	0.799			
Economic value (EV)	EV1	0.873	0.922	0.797	0.893
	EV2	0.946			
	EV3	0.857			
Relationship value (RV)	RV1	0.894	0.916	0.784	0.885
	RV2	0.854			
	RV3	0.908			
Dynamic capabilities (DC)	DC1	0.846	0.940	0.757	0.870
	DC2	0.844			
	DC3	0.872			
	DC4	0.902			
	DC5	0.886			

TABLE 4 (Continued)

Variable	Item	Normalized factor loading	Combination reliability	Average refined variance	Square root of average refined variance
Innovation capabilities (IC)	IC1	0.838	0.910	0.669	0.818
	IC2	0.811			
	IC3	0.901			
	IC4	0.781			
	IC5	0.752			
Competitive advantages (CA)	CA1	0.784	0.926	0.675	0.822
	CA2	0.792			
	CA3	0.853			
	CA4	0.820			
	CA5	0.877			
	CA6	0.799			
Innovation intelligibility (II)	II1	0.829	0.958	0.793	0.891
	II2	0.870			
	II3	0.895			
	II4	0.915			
	II5	0.921			
	II6	0.909			

5.3 | Structural equation model results

We reported the model fitting results, including $\chi^2/df = 2.75$, RMSEA = 0.048, CFI = 0.925, TLI = 0.905, and SRMR = 0.065. Therefore, the overall fit of the model is good. The structural equation model results are shown in Table 5. In summary, eight hypotheses are supported, and six hypotheses are not supported.

The structural equation model results show that the impact of value co-creation on economic value is positive ($\beta = 0.777$) with $p < 0.01$. The impact of value co-creation on relationship value is also positive ($\beta = 0.806$) with $p < 0.01$. Therefore, both Hypotheses 1 and 2 are supported. The results show that the impact of value co-creation on dynamic capabilities is positive ($\beta = 0.840$) with $p < 0.01$. Value co-creation positively correlates with innovation capability ($\beta = 0.652$) with $p < 0.01$. Therefore, both Hypotheses 3 and 4 are supported.

Given that the significance levels of the impact of economic value on the competitive advantage and innovation intelligibility are greater than 0.1 and the significance levels of the impact of the relationship value on the competitive advantage and innovation intelligibility are greater than 0.1, Hypotheses 5a, 5b, 6a, and 6b are not supported. The major reason may be that SMEs have flexibility and innovation. Consequently,

maintaining relationships with customers or obtaining economic value cannot significantly impact the company's competitive advantage and innovation intelligibility.

The structural equation model results show that the impact of dynamic capabilities on competitive advantage is positive ($\beta = 0.335$) with $p < 0.01$. The impact of dynamic capabilities on innovation intelligibility is also positive ($\beta = 0.416$) with $p < 0.01$. Therefore, both Hypotheses 7a and 7b are supported. The structural equation model results show that the influence of innovation capabilities on the competitive advantage is positive ($\beta = 0.355$) with $p < 0.01$. The influence of innovation capabilities on innovation intelligibility is also positive ($\beta = 0.377$) with $p < 0.01$. Therefore, both Hypotheses 8a and 8b are supported.

Considering that the significance level of the impact of value co-creation on competitive advantage and innovation intelligibility is greater than 0.1, both Hypotheses 9 and 10 are not supported. In fact, most existing research about value co-creation focus on large-scale enterprises in the context of digitalization. However, the data we collected are mainly from SMEs who usually convert needed knowledge to tacit innovation intelligibility and competitive advantage. Therefore, SMEs' participation in value co-creation cannot bring direct competitive advantage or innovation intelligibility.

TABLE 5 Result of structural equation model

Path	Standardization factor	P-value	Result
Value co-creation → economic value	0.777	0.000	Hypothesis 1 supported
Value co-creation → relationship value	0.806	0.000	Hypothesis 2 supported
Value co-creation → dynamic capabilities	0.840	0.000	Hypothesis 3 supported
Value co-creation → innovation capabilities	0.652	0.000	Hypothesis 4 supported
Economic value → competitive advantage	0.066	0.504	Hypothesis 5a not supported
Economic value → innovation intelligibility	−0.011	0.909	Hypothesis 5b not supported
Relationship value → competitive advantage	−0.076	0.526	Hypothesis 6a not supported
Relationship value → innovation intelligibility	−0.046	0.674	Hypothesis 6b not supported
Dynamic capabilities → competitive advantage	0.335	0.006	Hypothesis 7a supported
Dynamic capabilities → innovative intelligibility	0.416	0.001	Hypothesis 7b supported
Innovation capabilities → competitive advantage	0.355	0.001	Hypothesis 8a supported
Innovation capabilities → innovation intelligibility	0.377	0.000	Hypothesis 8b supported
Value co-creation → competitive advantage	0.240	0.185	Hypothesis 9 not supported
Value co-creation → innovation intelligibility	0.162	0.305	Hypothesis 10 not supported

TABLE 6 Result of mediating effect

Mediation effect	Standardized indirect effect estimation	95% confidence interval	
		Lower limit	Upper limit
Value co-creation → economic value → competitive advantage	$0.777 \times 0.066 = 0.051$	−0.097	0.200
Value co-creation → relationship value → competitive advantage	$0.806 \times (-0.076) = -0.061$	−0.251	0.128
Value co-creation → dynamic capabilities → competitive advantage	$0.84 \times 0.335 = 0.281$	0.088	0.475
Value co-creation → innovation capabilities → competitive advantage	$0.652 \times 0.355 = 0.231$	0.076	0.388
Value co-creation → economic value → innovation intelligibility	$0.777 \times (-0.011) = -0.008$	−0.158	0.140
Value co-creation → relationship value → innovation intelligibility	$0.806 \times (-0.046) = -0.037$	−0.211	0.137
Value co-creation → dynamic capabilities → innovation intelligibility	$0.84 \times 0.416 = 0.349$	0.133	0.566
Value co-creation → innovation capabilities → innovation intelligibility	$0.652 \times 0.377 = 0.246$	0.096	0.395

5.4 | Mediation effect

The Bootstrap method was used to test the mediating effect in this study. As shown in Table 6, with path coefficients 95% confidence interval, dynamic capabilities and

innovation capabilities have significant mediating effects between value co-creation and competitive advantage. Dynamic capabilities and innovation capabilities have significant mediating effects between value co-creation and innovation intelligibility.

6 | CONCLUSION

6.1 | Theoretical contribution and practical implication

We made several theoretical contributions. First, existing studies state that value co-creation can bring value to all parties. However, there are few systematic studies on this value composition. By comprehensively studying value co-creation, this research can bring various values, including explicit value (economic value and relationship value) and implicit value (dynamic capabilities and innovation capabilities), and analyse the impact of related value on competitive advantage and innovation intelligibility. Second, this research investigates the influencing mechanism between value co-creation, associated value, competitive advantage, and innovation intelligibility from the AI innovation ecosystem perspective and reveals the mediating effect of dynamic capabilities and innovation capabilities. Third, maintaining competitive advantage and innovation intelligibility for SMEs in the digital era has attracted attention from researchers. This research focuses on SMEs and identifies the mechanism for them to maintain competitive advantage and improve innovation intelligibility from the AI innovation ecosystem perspective.

This research has several practical implications, especially for SMEs. First, SMEs should actively participate in the value co-creation activities of the AI innovation ecosystem. It is difficult for SMEs to have all the resources needed for digital transformation. The cooperation with large enterprises, universities, research institutions, and governments can enable SMEs to realize the complementary advantages from other collaborators and promote digital transformation and capacity improvement (Mingxing et al., 2020; Yang et al., 2018). Second, SMEs should focus on the cultivation of dynamic capabilities and innovation capabilities. Knowledge and resources flow from large companies to SMEs in the AI innovation ecosystem. Knowledge is transformed into innovation output. The knowledge spillover effect will occur in the value co-creation process. SMEs are usually good at a certain professional field of knowledge. Learning and absorbing the spilled knowledge from collaborators is essential for SMEs' competitive advantage and capacity improvement.

6.2 | Limitation and future research

This research has several limitations. First, the AI innovation ecosystem era has new characteristics. Specifically, this research did not consider enterprises' digital level

(Bondar et al., 2017; Li et al., 2008, 2018; Priyanka et al., 2021; Wang et al., 2007; Xu, 2013, 2020; Xu et al. 2005, 2012a,b; Xu et al., 2013; Xu & Viriyasitavat, 2014, 2019), which has a certain impact on value co-creation. Thus, one possible future research can use the digital level of enterprises as a control variable to examine its impact on enterprises. Second, this research mainly focuses on the AI-related manufacturers' production and consumption processes. Thus, a more comprehensive research framework could be proposed to explore the value co-creation in each process and the multi-dimensional value acquired by the enterprise and the customer. Third, we collected the data from the Yangtze River Delta region, where SMEs are relatively concentrated. The economic development level and the cultural background of this region are different from those of other regions in China. Therefore, more data could be collected from enterprises located in other regions for further study and insightful comparison. Last, the value co-creation mechanism of the AI innovation ecosystem has played an important role in value creation and value acquisition. We may investigate value co-creation's operation and governance mechanism and its impact on system evolution.

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DATA AVAILABILITY STATEMENT

Due to privacy concerns, neither the data nor the source of the data can be made available.

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